EX NO: 29

SIMULATION AND ANALYSIS OF GSM COMMUNICATION SYSTEM WITH TIME DIVISION MULTIPLE ACCESS (TDMA) USING MATLAB

Objectives

1. To simulate the operation of a GSM communication system using Time Division Multiple To understand the basic structure of a GSM communication system and the role of Time Division Multiple Access (TDMA).
2. To generate a digital information sequence and organize it into GSM-style time slots using MATLAB.
3. To construct a simplified GSM/TDMA frame by assigning users to different time slots.
4. To simulate transmission, reception, slot extraction and recovery of the intended user data.
5. To observe the effect of slot allocation on multiple-access operation and verify that users can share the same carrier in different time slots.
6. To calculate and analyze important TDMA parameters such as number of slots, slot duration, frame duration and duty cycle.

CODE

```
clc;
clear;
close all;

% Data Bits
data = [1 0 1 1 0];

% Convert binary data to +1 and -1
data_bipolar = 2*data - 1;

% PN Sequence (Length = 5)
pn = [1 -1 1 -1 1];

% DSSS Spreading
spread = kron(data_bipolar, pn);

% Despreading
rec = reshape(spread, length(pn), []);
rec_data_bipolar = sign(sum(rec, 1));

% Convert back to binary
rec_data = (rec_data_bipolar + 1)/2;

% Display Results
disp('Original Data:');
```

```
disp(data);

disp('PN Sequence:');
disp(pn);

disp('DSSS Spread Signal:');
disp(spread);

disp('Recovered Data:');
disp(rec_data);

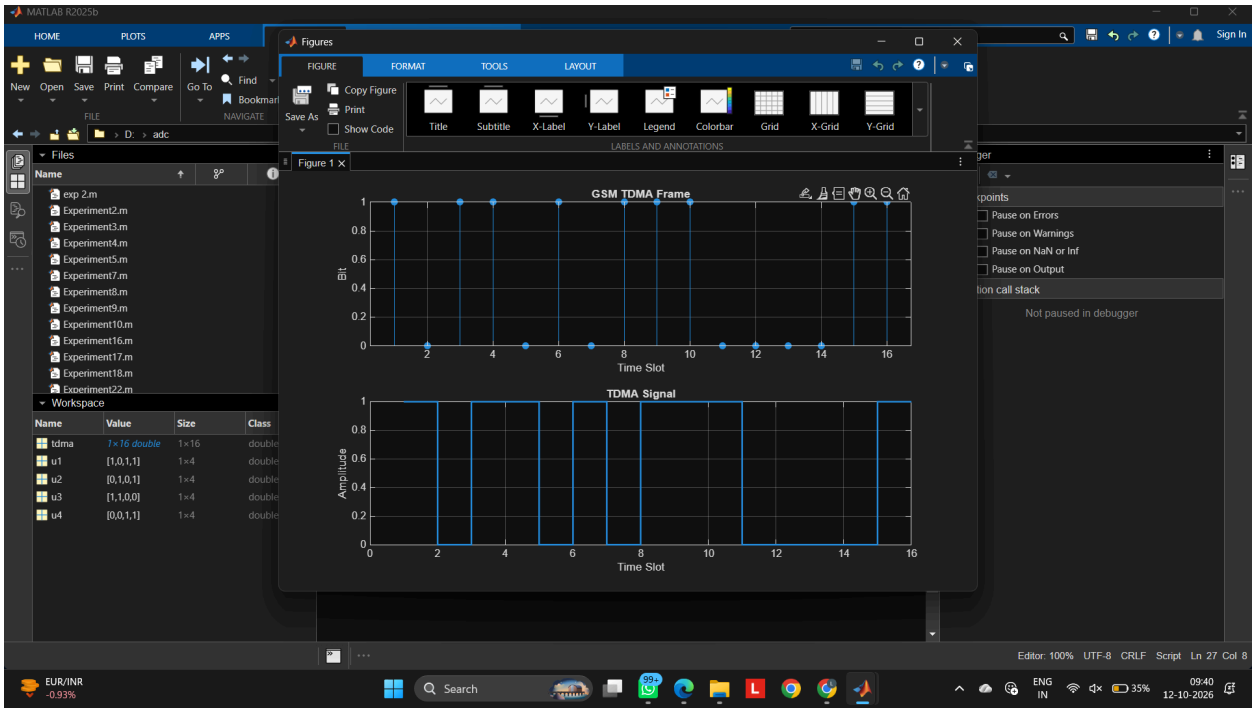
% Plots
figure('Name','DSSS Spreading and Despreading', ...
      'NumberTitle','off');

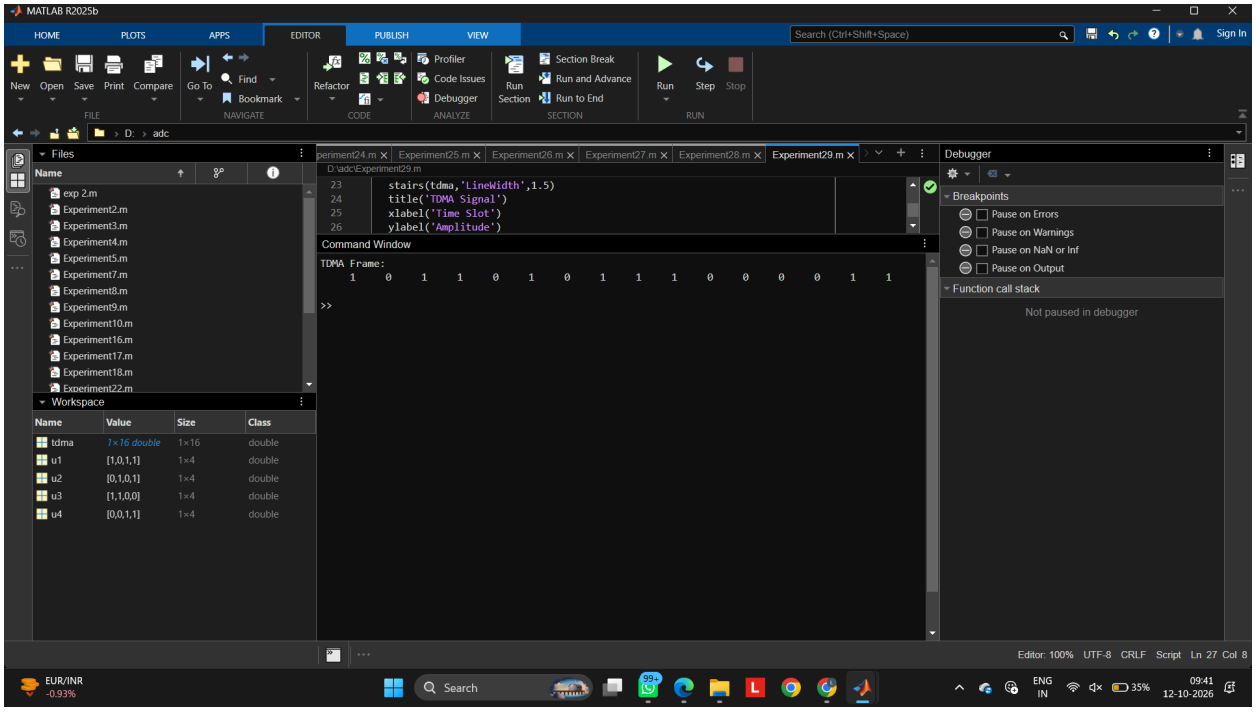
subplot(3,1,1);
stem(0:length(data)-1, data, 'filled');
title('Original Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(3,1,2);
stem(0:length(spread)-1, spread, 'filled');
title('DSSS Spread Signal');
xlabel('Chip Index');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
stem(0:length(rec_data)-1, rec_data, 'filled');
title('Recovered Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
```

OUTPUT


GSM TDMA Frame
Bit
Time Slot
TDMA Signal
Amplitude
Time Slot


TDMA Frame:
1 0 1 1 0 1 0 1 1 1 0 0 0 0 1 1
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